

Xinyu Huang

Shanghai, China | contact@x3huang.dev | x3huang.dev | github.com/XyLearningProgramming

TECHNICAL HIGHLIGHTS

AI Product Specialist: 5 years' experience delivering intelligent speech SaaS solutions; expert in ML model operationalization (feature engineering, deployment, and monitoring).

Cloud-Native Architect: Expert in microservices andDDD; proficient in HTTP, Websocket, gRPC/Thrift, and GraphQL.

High-Performance Systems: Built gateways supporting 1M+ QPS with dynamic rate-limiting and multi-layer caching.

DevOps Practitioner: Familiar with CI/CD tools (Jenkins, ArgoCD) and gray-scale deployment strategies.

EXPERIENCE

Rakuten

Aug. 2025 – Present

Senior Software Engineer, Agent Experience, Rakuten AI

Shanghai, China

- Architected a reactive agent framework and microservice refactorization for real-time audio interactions by leveraging **Project Reactor** and Spring Cloud to manage multi-modal data streams with backpressure; achieved a **10x reduction in CPU usage** and sustained **10k+ DAU** for real-time agents.
- Enhanced real-time agent cognition by integrating a custom memory module (semantic RAG) and a dynamic LLM Routing layer; implemented sub-agent handoff graphs that delegated specialized tasks, resulting in a significantly improved response relevance for specialized agents given custom evaluation sets.
- Standardized the AI lifecycle by building an automated **LLM-as-a-Judge** evaluation pipeline with other WER, BLEU, ROUGE metrics and a GKE-based canary deployment strategy with Istio and GrowthBook; ensured **99% stability** of complex audio pipelines across frequent model updates and mobile client versions.

Freelancer

May 2025 – Aug. 2025

Full-Stack Developer

Remote

- Developed a personal technical blog and (x3huang.dev) using a self-hosted k3s cluster to implement OTLP-based monitoring (Prometheus/Grafana) and interactive Unity WebGL modules.

Bytedance, Inc.

Jan. 2021 – Apr. 2025

Backend Software Engineer, Platform Infrastructure, Audio and Speech

Shanghai, China

- Engineered the high-concurrency speech gateway for **Doubao (LLM)** real-time mode and VolcEngine SaaS (Task), utilizing Raft-based consensus and dynamic load balancing to sustain **99.99% availability** for a global traffic load exceeding **10B+ daily requests**.
- Scaled the BytePlus enterprise platform by integrating multi-tenant JWT/STS authentication and complex pricing engines; these robust security and billing frameworks facilitated a **100x growth** in the ToB client base while maintaining milliseconds authentication overhead.
- Optimized the multi-modal data feedback loop by developing a Flink and Spark-based ETL pipeline processing **300TB of daily speech data**, directly accelerating model training cycles and improving the accuracy of proprietary ASR/TTS iterations.

San Diego Hunger Coalition (NGO)

Jul. 2019 – Aug. 2019

Data Intern

San Diego, US

- Automated the analysis of 500k+ regional food security records by engineering Python-based ETL scripts and optimizing Amazon RDS star-schema architectures to streamline non-profit reporting.

EDUCATION

University of California, San Diego

San Diego, US

Master of International Affairs (Core: Quantitative Methods, Statistical Learning)

Sept. 2018 – June 2020

- Capstone: Do People Divorce Because of a Policy? Impact of Limit Purchasing Orders in China.
- ML Project: On Automatic X-ray Images Generation from Radiology Reports.

Shanghai International Studies University

Shanghai, China

Bachelor of Arts in Translation and Interpretation

Sept. 2014 – June 2018

黄昕宇

中国上海 | contact@x3huang.dev | x3huang.dev | github.com/XyLearningProgramming

技术亮点

- AI 产品专家:** 5 年智能语音SaaS 交付经验, 深耕大模型Agent 架构设计与模型服务化 (特征工程、全链路部署与可观测性监控)。
- 云原生架构师:** 精通微服务架构设计与领域驱动设计 (DDD); 深入理解HTTP, Websocket, gRPC/Thrift 及GraphQL 协议。
- 高性能系统:** 具备百万级QPS 网关开发经验, 擅长设计动态限流、分布式共识 (Raft) 及多级缓存策略。
- DevOps 专家:** 熟练运用CI/CD 工具链 (Jenkins, ArgoCD) 及GKE 灰度发布策略, 保障AI 业务管线的高可用性。

工作经验

- Rakuten** 2025年8月– 至今
高级软件工程师, *Agent Experience, Rakuten AI* 中国上海
- 主导Rakuten AI 实时语音模式的响应式微服务重构, 基于**Project Reactor** 与Spring Cloud 构建多模态流处理框架, 引入背压 (Backpressure) 机制优化数据流, 实现**10倍CPU 性能提升** 并支撑数万日活用户。
 - 提升智能体认知能力与成本效能: 引入自定义用户记忆模块 (基于语义RAG) 及LLM 路由层; 设计子智能体手接管图 (Handoff Graphs), 将特定任务分配至最优模型方案, 最终显著增强agents在特定场景的返回相关性。
 - 标准化AI 交付生命周期: 构建基于**LLM-as-a-Judge**的自动化评测管线, 同时整合WER, BLEU, ROUGE 等指标; 利用GKE Canary、Istio 流量治理及GrowthBook 实现多版本移动端及复杂音频管线的灰度发布, 保障系统**99.9%** 稳定性。
- 自由职业** 2025年5月– 2025年8月
全栈开发工程师 远程
- 基于自托管k3s 集群开发并运维个人技术博客(x3huang.dev), 集成OTLP 标准的可观测性方案 (Prometheus/Grafana) 及Unity WebGL 交互模块。
- 字节跳动(Bytedance, Inc.)** 2021年1月– 2025年4月
后端开发工程师, 平台基础设施 (语音团队) 中国上海
- 参与**豆包(Doubao)** 大模型实时通话 及火山引擎语音网关建设, 利用Raft 共识算法设计高性能负载均衡系统, 在高并发场景下保障了**每日百亿级请求的99.99%** 高可用性。
 - 驱动BytePlus 海外企业级平台规模化增长: 集成多租户JWT/STS 认证与复杂计费引擎, 支撑ToB 客户量实现**百倍增长**, 同时将认证延迟控制在毫秒级。
 - 优化多模态数据反馈闭环: 利用Flink 与Spark 构建海量数据ETL 管线 (日处理量**300TB**), 通过定价数据与训练语料的实时处理, 直接加速了自研ASR/TTS 模型的迭代效率。
- San Diego Hunger Coalition (NGO)** 2019年7月– 2019年8月
数据科学实习生 美国圣地亚哥
- 通过Python 自动化处理50 万余条区域粮食安全记录, 优化Amazon RDS 星型模型架构, 提升非营利组织业务报告生成效率。

教育背景

- 加州大学圣地亚哥分校(UCSD)** 美国圣地亚哥
国际事务硕士 (核心方向: 定量分析、统计学习) 2018年9月– 2020年6月
- 毕业设计: 政策对离婚率的影响研究 (中国限购政策准实验分析)。
 - 机器学习项目: 基于放射科报告的X 光图像自动生成研究。
- 上海外国语大学** 中国上海
文学学士 (英语翻译专业) 2014年9月– 2018年6月

项目与技能

- My_wacig** | *Golang*, 编译器设计 2022年10月
- 基于Monkey 语言开发轻量级编译器, 扩展了自定义类型系统及语法特性, 深入理解程序语言编译原理。
- 编程语言:** Golang, Python, Java, C++, SQL, R
- 工具与框架:** LangGraph, OpenAI Agents SDK, Project Reactor, Spring AI, Docker, Kubernetes (k3s/GKE), Helm, Istio, Flink, Spark, Git